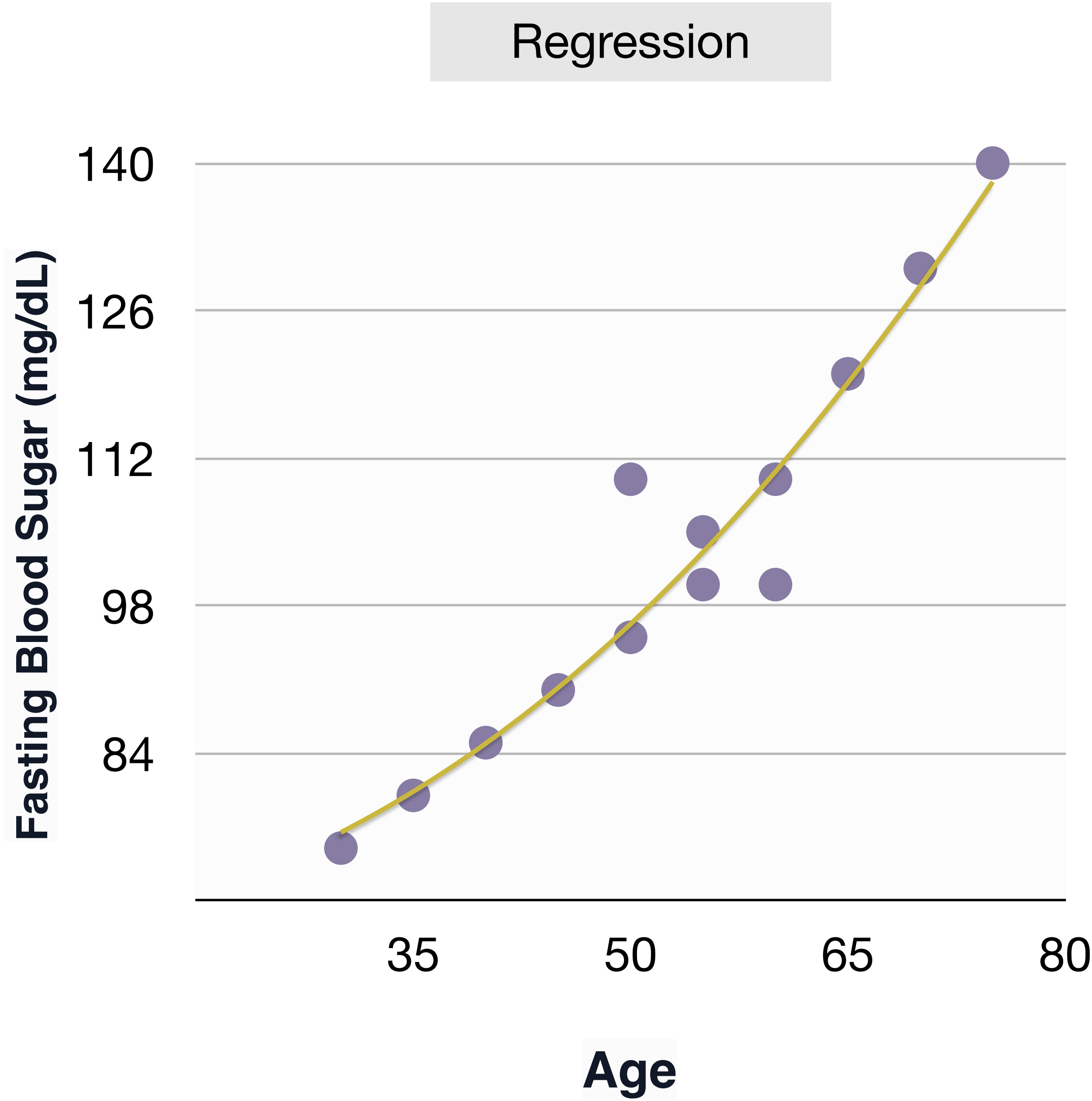
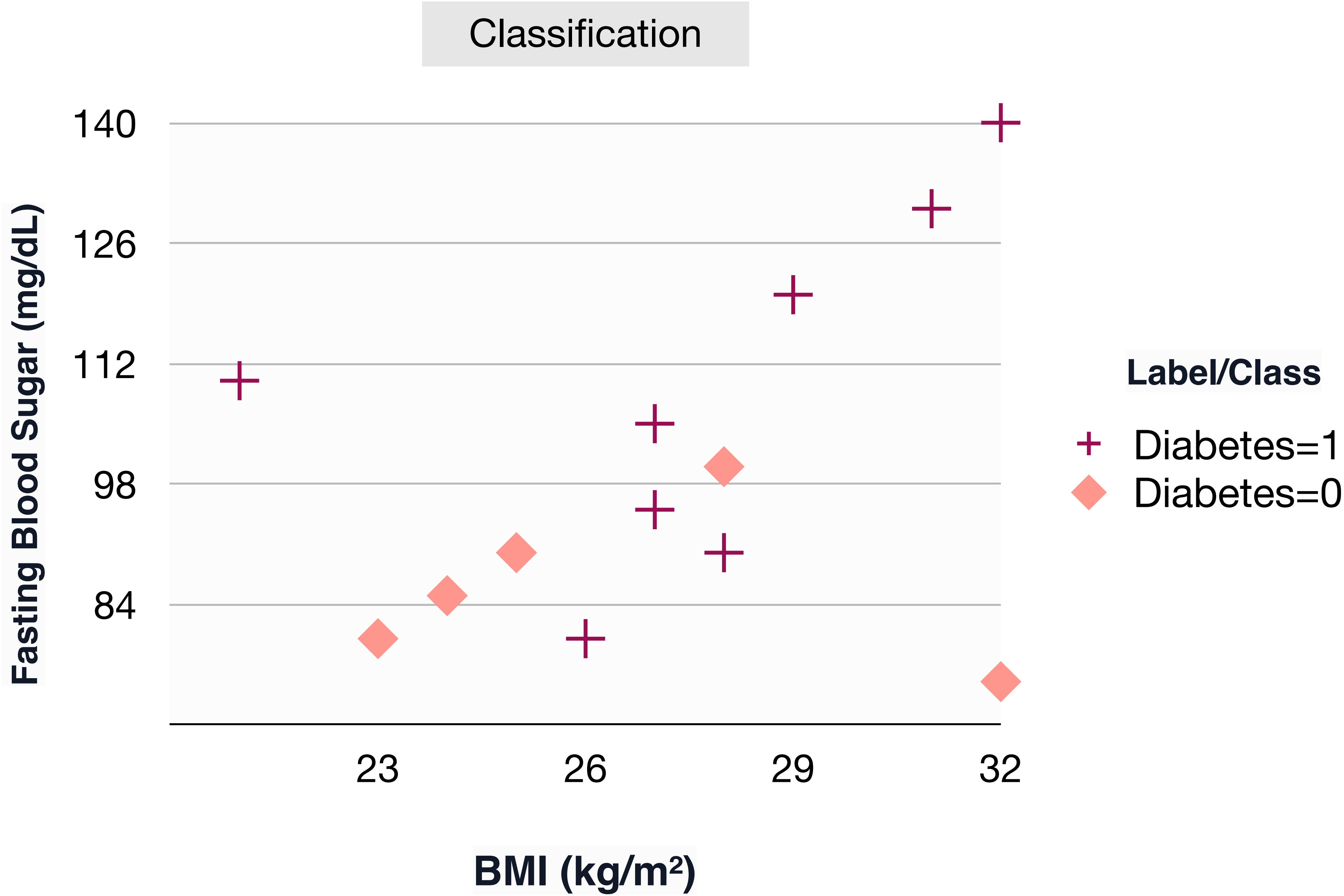
SUPERVISED LEARNING

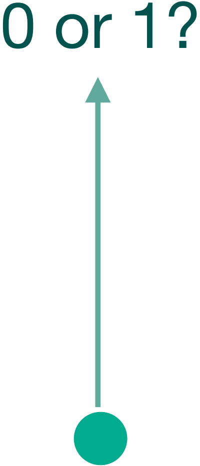
What Kind of Labels?

The dataset is a collection of tuples: $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \ldots, (x^{(n)}, y^{(n)})\}$ Where:

- $x^{(i)} = [x_1^{(i)}, x_2^{(i)}, \ldots, x_m^{(i)}]$ is a vector of features for the i^{th} instance.
  - $y^{(i)}$ is the continuous target variable for the i^{th} instance [Regression].
  - $y^{(i)}$ is the categorical (discrete) class label for the i^{th} instance [Classification].


Regression
140
126
112
98
84
Fasting Blood Sugar (mg/dL)
35
50
65
80
Age


Classification
Fasting Blood Sugar (mg/dL)
140
126
112
98
84
23
26
29
32
BMI (kg/m²)
Label/Class
Diabetes=1
Diabetes=0


0 or 1?

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SUPERVISED LEARNING

What Kind of Labels?

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  - $y^{(i)}$ is the continuous target variable for the i^{th} instance [Regression].
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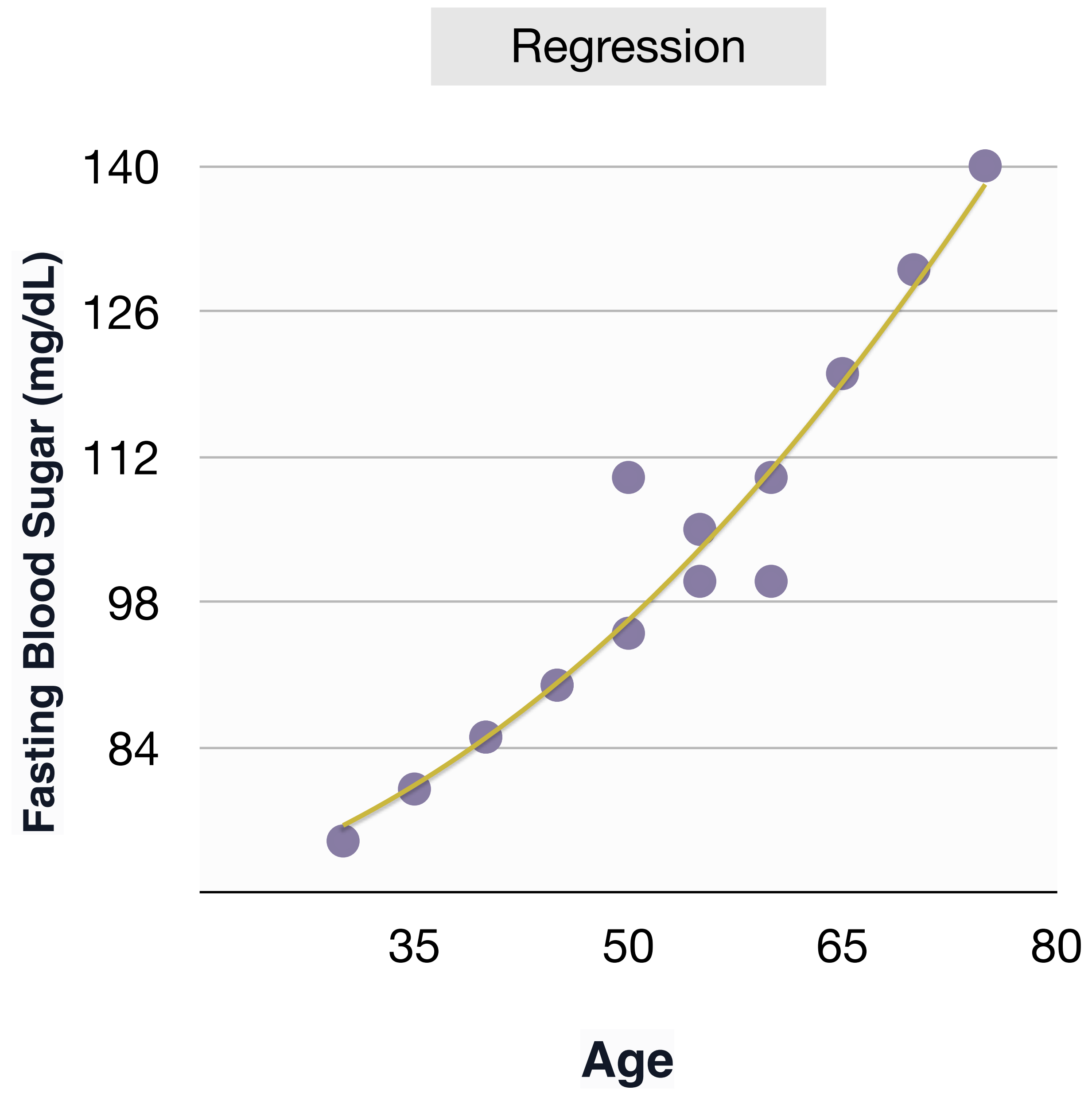
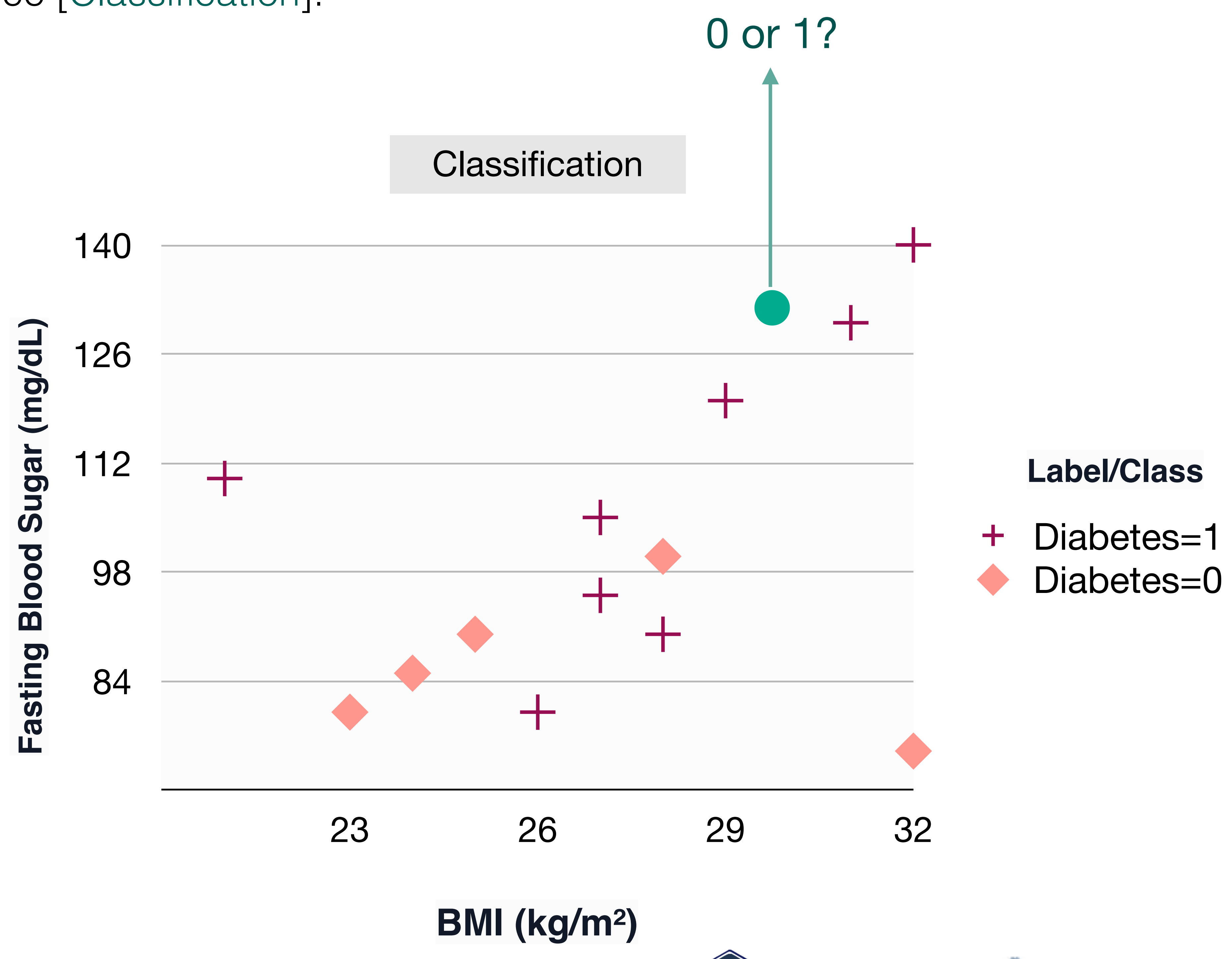




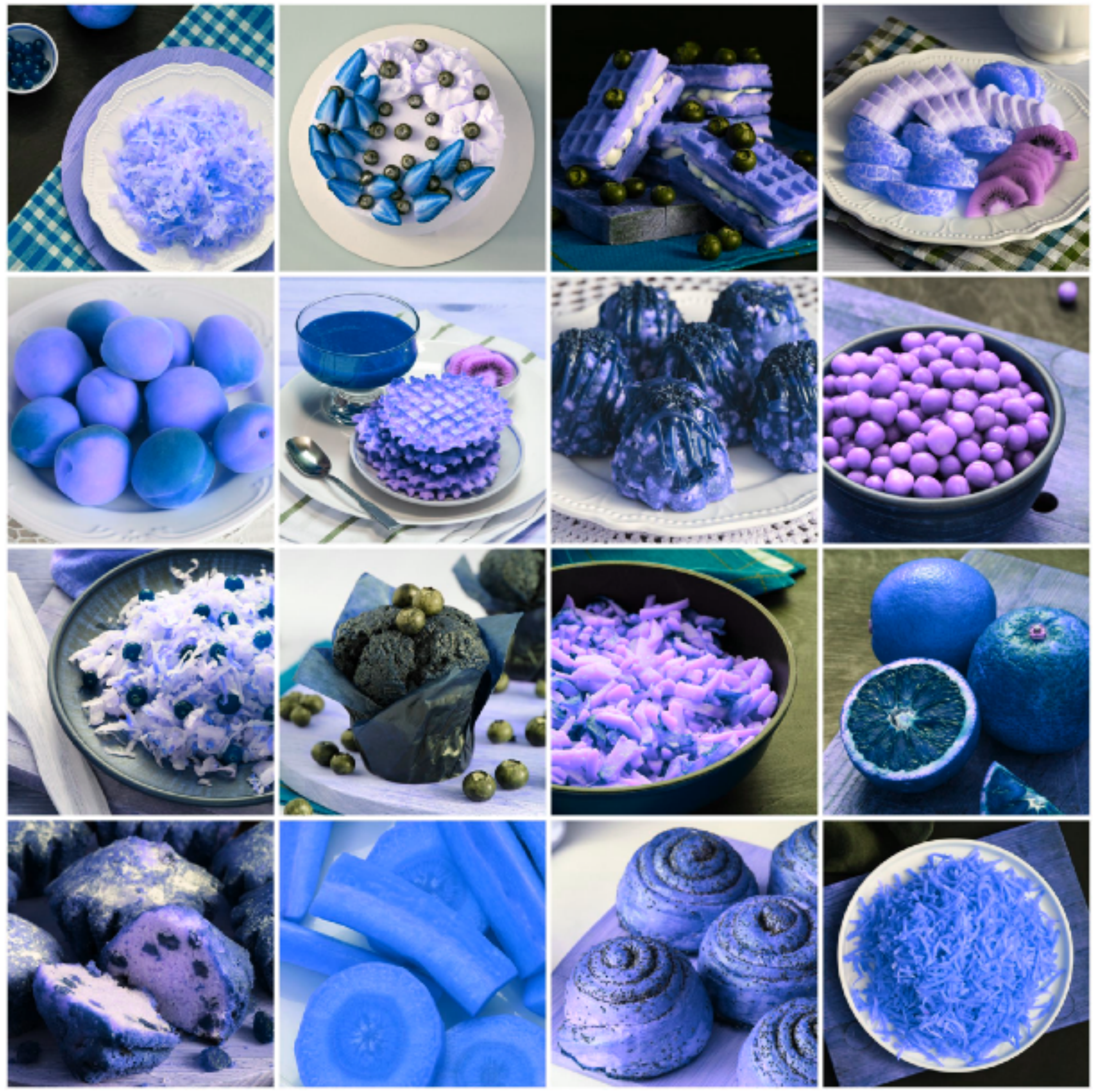
SUPERVISED LEARNING

What Kind of Data Types?

The dataset is a collection of tuples:

$\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \ldots, (x^{(n)}, y^{(n)})\}$ Where:

- $x^{(i)} = [x_1^{(i)}, x_2^{(i)}, \ldots, x_m^{(i)}]$ is the numerical representation of the image for classification. It is created by flattening pixel values into a 1D array. For instance, a 100x100 pixel image yields a 10,000-element grayscale vector or a 30,000-element RGB vector.
- $y =$ The probabilities of x for each class/category.



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